ATHAR HUSSAIN

Data Scientist & Machine Learning Engineer
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PROFESSIONAL SUMMARY

Aspiring Data Scientist and Machine Learning Engineer with hands-on experience building end-to-end Machine Learning, Deep Learning, and AI applications. Strong foundation in Python, SQL, Scikit-learn, PyTorch, RAG, model evaluation, feature engineering, hyperparameter optimization, and deployment. Built 10 projects spanning predictive modeling, computer vision, conversational RAG, customer analytics, and relational database design, with demonstrated results including 0.848 ROC-AUC, 0.823 R^2, and 77.22% test accuracy. B.Tech CSE student with 9+ SGPA focused on practical, reproducible, and production-oriented AI systems.

TECHNICAL SKILLS

Programming: Python, OOP, SQL, Pandas, NumPy
Machine Learning: Scikit-learn, Linear/Logistic Regression, Decision Trees, Random Forest, AdaBoost, Gradient Boosting, XGBoost, K-Means
Deep Learning: PyTorch, ANN, CNN, Torchvision, Dataset/DataLoader, Batch Normalization, Dropout, Early Stopping, Data Augmentation
AI / RAG: RAG, FAISS, Sentence Transformers, Embeddings, Groq LLM, LangChain, Conversational Question Rewriting
Model Development: Feature Engineering, Pipeline, ColumnTransformer, Cross-Validation, GridSearchCV, RandomizedSearchCV, Optuna, SHAP
Evaluation: Accuracy, Precision, Recall, F1-Score, ROC-AUC, R^2, MAE, RMSE, Confusion Matrix, Threshold Optimization
Deployment & Tools: Streamlit, Streamlit Community Cloud, Git, GitHub, Joblib, Jupyter, VS Code, Google Colab
Databases: SQL Server, SSMS, Relational Database Design

PROJECTS

Machine Learning RAG Assistant

Tech: Python | RAG | FAISS | Sentence Transformers | Groq LLM | LangChain | Streamlit

- Built a conversational RAG application for querying a custom Machine Learning knowledge base. Implemented semantic retrieval with FAISS and Sentence Transformer embeddings, conversational question rewriting, grounded LLM generation, source-topic display, and out-of-knowledge-base handling.
  LIVE APPLICATION: athars-ml-rag-application.streamlit.app
  GitHub: github.com/ATHAR-HUSSAIN-2910/machine-learning-rag

Diabetes Risk Prediction with ANN

Tech: PyTorch | ANN | Optuna | Scikit-learn | Streamlit

- Developed a feed-forward ANN on 70,692 BRFSS records with leakage-safe preprocessing and stratified 70/15/15 splits. Optimized architecture and training hyperparameters with Optuna, applied early stopping and validation-based threshold optimization; achieved 0.830 ROC-AUC, 89.4% recall, and 77.6% F1 on the untouched test set.
  LIVE APPLICATION: athars-diabetes-ann-pytorch.streamlit.app
  GitHub: github.com/ATHAR-HUSSAIN-2910/diabetes-ann-pytorch

CIFAR-10 Image Classification with Custom CNN

Tech: PyTorch | CNN | Computer Vision | Torchvision | Optuna

- Designed a custom CNN using Conv-BatchNorm-ReLU-MaxPool blocks for 60,000 32×32 RGB images. Applied training-only augmentation and Optuna tuning; achieved 77.22% test accuracy with 0.772 macro F1 and performed confusion-matrix and class-level error analysis.
  GitHub: github.com/ATHAR-HUSSAIN-2910/cifar10-cnn-pytorch

Customer Churn Prediction

Tech: Python | XGBoost | Scikit-learn | SHAP | Streamlit

- Compared six classification models and tuned XGBoost with GridSearchCV for telecom churn prediction. Achieved 79.7% accuracy and 0.848 ROC-AUC, applied SHAP for model explainability, and deployed the best model as an interactive application.
  LIVE APPLICATION: athars-customer-churn-prediction.streamlit.app
  GitHub: github.com/ATHAR-HUSSAIN-2910/Customer-Churn-Prediction

Customer Order SQL Project – Database Design & Analytics

Tech: SQL Server | SSMS | Python | Pandas | Faker

- Designed a relational e-commerce database covering 200 customers, 40 products, and 1,484 orders. Generated realistic synthetic data with Python and authored 85 SQL queries using JOINs, CTEs, Window Functions, and aggregations to answer business questions.
  GitHub: github.com/ATHAR-HUSSAIN-2910/Customer-Order-SQL-Project

House Price Prediction

Tech: Python | Scikit-learn | Pipeline | ColumnTransformer | Streamlit

- Built Linear, Ridge, Lasso, and Elastic Net regression pipelines with feature engineering, cross-validation, and hyperparameter tuning. Achieved $R^2 = 0.823$ with Lasso Regression and deployed the model for real-time predictions.
  LIVE APPLICATION: athars-house-price-prediction.streamlit.app
  GitHub: github.com/ATHAR-HUSSAIN-2910/house_price_prediction

Titanic: Comparative Classification Study

Tech: Python | Scikit-learn | Pipeline | GridSearchCV | Streamlit

- Built an end-to-end classification pipeline comparing Logistic Regression, Decision Tree, Bagging, and Random Forest. Applied Pipeline/ColumnTransformer preprocessing and GridSearchCV, benchmarked multiple classification metrics, and deployed the top model.
  GitHub: github.com/ATHAR-HUSSAIN-2910/titanic-survival-model-comparison

EDUCATION

B.Tech in Computer Science Engineering (CSE) — Shadan College of Engineering and Technology, Hyderabad | Expected Graduation: 2027 | SGPA: 9+ across 6 semesters

ADDITIONAL INFORMATION

Profile: Open to Data Science / Machine Learning internships and fresher roles | Deployment: 4 Streamlit Community Cloud applications | Focus: Machine Learning, Deep Learning, AI Engineering, RAG